

A PROJECT REPORT
ON
“LOAN PREDICTION MODEL USING MACHINE LEARNING &
CREDIT HEALTH PREDICTION SYSTEM
”

BACHELOR OF TECHNOLOGY
IN
AI&ML/ CYBER SECURITY



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CANDIDATE'S DECLARATION

I hereby declare that this project report titled "**[LOAN PREDICTION MODEL USING ML AND CREDIT HEALTH PREDICTION SYSTEM]**" is an original work done by me under the supervision of **[Mr. Manoj Kumar]**. It has not been submitted previously for the award of any degree.

Date: 12-05-26

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CERTIFICATE

This is to certify that the project titled "[**LOAN PREDICTION MODEL USING ML AND CREDIT HEALTH PREDICTION SYSTEM**]" submitted by [**Ananya Kapil**] Roll no-
[**243022029**], has been carried out under my guidance and is approved for submission.

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[Ananya Kapil]

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ABSTRACT

The **AI-Powered Loan Approval and Credit Health Prediction System** is a machine learning-based web application developed to improve traditional loan approval analysis by considering both financial and behavioral factors. Conventional loan approval systems often depend heavily on the CIBIL score alone, which may lead to biased or incomplete decisions. This project aims to create a smarter and more transparent system that evaluates a user's overall financial profile before predicting loan approval. The system uses various machine learning algorithms such as Logistic Regression, Random Forest, Gradient Boosting, XGBoost, SVM, and KNN to analyze applicant data including annual income, loan amount, assets, employment status, dependents, and CIBIL score. In addition to these traditional parameters, new credit behavior features such as missed EMIs, credit utilization, active loans, credit history length, and recent credit inquiries were integrated to improve prediction quality and improves credit score. A user-friendly web interface was developed using Flask and HTML/CSS, allowing users to input financial details and receive instant loan approval predictions along with a visual credit health indicator and personalized suggestions for improving their credit profile. The system not only predicts loan eligibility but also acts as a decision-support tool by providing actionable financial guidance.

INTRODUCTION

today's digital financial ecosystem, banks and financial institutions use automated systems to evaluate whether a customer is eligible for a loan. Most traditional loan approval systems primarily depend on the applicant's **CIBIL score** and basic financial details. However, relying heavily on a single parameter may not always provide an accurate understanding of a person's financial reliability and repayment capability. The **AI-Powered Loan Approval and Credit Health Prediction System** was developed to create a smarter and more balanced loan evaluation process using machine learning techniques. The project aims to analyze not only standard financial parameters such as annual income, loan amount, assets, employment status, and education level, but also important credit behavior factors including missed EMI payments, credit utilization, active loans, credit history duration, and recent credit inquiries. The system uses multiple machine learning algorithms to predict whether a loan is likely to be approved or rejected based on the applicant's overall financial profile. In addition to loan prediction, the project also introduces a **Credit Health Analysis** feature that visually represents the applicant's credit condition and provides personalized suggestions to improve their financial profile and CIBIL score.

LITERATURE REVIEW

- Machine learning models improve accuracy in loan approval and credit risk prediction.
- Studies on financial datasets show the impact of:
 - Applicant income
 - Loan amount
 - CIBIL score
 - Employment status
 - Asset ownership on loan approval decisions
- Recent research highlights the importance of behavioural credit factors such as:
 - Missed EMI payments
 - Credit utilization ratio
 - Credit history duration
 - Active loans and recent credit inquiries
- Common prediction methods include:
 - Logistic Regression
 - Random Forest
 - Gradient Boosting
 - XGBoost
 - Support Vector Machine (SVM)
 - K-Nearest Neighbours (KNN)
- Goal: Provide intelligent loan prediction and practical credit improvement support for users.

SYSTEM ANALYSIS

9.1 Existing System

The proposed system is an AI-based Loan Approval and Credit Health Prediction application designed to improve traditional loan assessment methods. Existing loan approval systems mainly depend on CIBIL scores and limited financial parameters, which may result in biased or incomplete decisions. The system analyzes multiple financial and behavioral attributes to provide more intelligent and transparent predictions.

9.2 PROPOSED SYSTEM

The application accepts user inputs such as income, loan amount, assets, employment status, dependents, CIBIL score, missed EMIs, credit utilization, active loans, credit history, and recent credit inquiries. These inputs are processed using machine learning models to predict loan approval status and generate a credit health evaluation.

SYSTEM DESIGN

10.1 ARTITECTURE DIAGRAM

The system is designed using a machine learning-based web application architecture consisting of frontend, backend, and prediction modules. The frontend interface is developed using HTML and CSS, where users can enter their financial and personal information through input forms. The backend is developed using Flask, which handles data processing, model execution, and result generation.

10.2 DFD (DATA FLOW DIAGRAM)

The Data Flow Diagram represents the movement of data within the system. Initially, the user provides personal, financial, asset, and credit behavior details through the web interface. The entered data is transferred to the Flask backend, where preprocessing and validation operations are performed. The processed data is then passed to the trained machine learning model for prediction. The model analyzes the input features and generates loan approval status along with credit health analysis. Finally, the output data containing prediction results and

10.3 ENTITY -RELATIONSHIP DIAGRAM

The ER Diagram describes the relationship between different entities involved in the system. The primary entity is the User, who provides multiple attributes such as personal information, financial details, asset information, and credit behavior data. These attributes are connected to the Prediction entity, which stores the loan approval result and credit health score generated by the machine learning model. Another entity, Suggestions, is associated with the prediction results and contains personalized recommendations for improving credit score and financial behavior.

11.1 Technologies Used & 11.2 Coding and Modules

Python

Python is the primary programming language used in this project. It is widely used in machine learning and data science because of its simplicity, readability, and extensive library support. Python was used for data preprocessing, model training, backend development, and application integration.

Flask

Flask is a lightweight Python web framework used to develop the backend of the web application. It handles routing, form processing, prediction requests, and communication between the frontend and the machine learning model.

HTML

HTML (HyperText Markup Language) is used to create the structure of the web pages. It defines the layout of forms, buttons, input fields, and result sections in the application interface.

Pandas

Pandas is a Python library used for data manipulation and analysis. It was used for reading datasets, handling missing values, preprocessing data, and converting user inputs into structured formats for machine learning models.

Scikit-learn

Scikit-learn is a machine learning library used for building and evaluating prediction models. Algorithms such as Logistic Regression, Random Forest, Gradient Boosting, SVM, and KNN were implemented using this library.

XGBoost

XGBoost is an advanced machine learning algorithm used for high-performance classification tasks. It improves prediction accuracy and efficiency through optimized gradient boosting techniques.

Joblib

Joblib is used for saving and loading trained machine learning models. It allows the trained model to be reused in the Flask application without retraining every time the application runs.

VS Code

Visual Studio Code (VS Code) is the code editor used for project development. It provides features such as debugging, terminal support, extensions, and syntax highlighting for efficient coding and testing.

TESTING AND VALIDATION

System validation was performed to ensure that the Loan Approval and Credit Health Prediction System functions accurately and reliably. Different test cases were used by entering various combinations of financial, personal, and credit behavior data into the application. The system successfully generated loan approval predictions, calculated credit health scores, and provided personalized suggestions based on user input. Multiple machine learning models were evaluated and compared using accuracy metrics and classification reports to validate prediction performance. The web application was also tested for proper input handling, responsive user interface behavior, and smooth integration between the frontend, backend, and machine learning modules.

RESULTS & DISCUSSION

The key outcomes of the project include:

- Successful training and deployment of multiple machine learning models for loan approval prediction.
- Accurate loan approval and rejection predictions generated through the Flaskbased web application.
- Integration of additional credit behaviour parameters reduced excessive dependency on only the CIBIL score.
- Credit Health Score successfully generated based on financial and behavioural analysis.
- Personalized suggestions provided to help users improve their credit profile and financial habits.
- Visual credit health indicators improved interpretability and user understanding of prediction results.
- Successful implementation of scenario-based testing by changing financial and behavioural attributes to observe prediction outcomes.

CONCLUSION

This project successfully developed an AI-powered Loan Approval and Credit Health Prediction System using machine learning techniques. The system analyses both financial and behavioural factors to provide more balanced and intelligent loan approval decisions. Unlike traditional systems that rely mainly on the CIBIL score, this project integrates additional parameters such as missed EMIs, credit utilization, active loans, and credit history. Multiple machine learning models were trained and evaluated to achieve accurate predictions. A user-friendly Flask web application was created to provide real-time interaction and prediction results. The system also generates a Credit Health Score and personalized suggestions for improving financial behaviour. Visual indicators and interactive UI components improved transparency and user understanding. The project demonstrates the practical application of machine learning in financial decision-making systems. It also highlights the importance of behavioural analysis in credit evaluation. Overall, the system transforms a simple prediction model into an intelligent financial decision-support platform.

FUTURE WORK

Future enhancements can significantly expand the system's capabilities:

- Integration of advanced deep learning models for improved financial and credit risk analysis.
- Real-time financial data integration through banking APIs and live credit monitoring systems.
- Cloud deployment for better scalability, security, and multi-user accessibility
- Development of mobile and responsive web applications for wider usability and accessibility.
- Automated financial report generation and intelligent credit analysis summaries.
- Addition of AI-based financial assistant features for personalized loan and credit guidance.
- Enhanced visualization dashboards with interactive graphs, charts, and credit analytics.
- Integration with banking and financial institution systems for real-world loan processing applications.
- Implementation of fraud detection and risk analysis modules for safer loan approval processes.

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